1.

$$\begin{pmatrix} 4+c_2 & 1+c_2 \\ 1+c_2 & 1 \end{pmatrix}\begin{pmatrix} \ddot{\theta}_1 \\ \ddot{\theta}_2 \end{pmatrix} + \begin{pmatrix} -(2\dot{\theta}_1\dot{\theta}_2+\dot{\theta}_2^2)s_2 \\ \dot{\theta}_1^2 s_2 \end{pmatrix} = \begin{pmatrix} \tau_1 \\ \tau_2 \end{pmatrix}$$

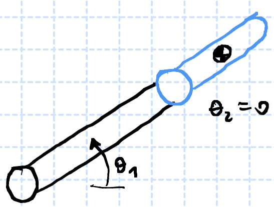
Joint 2 is locked, $\theta_2 = \theta_2^*$. Therefore $\ddot{\theta}_2 = \dot{\theta}_2 = 0$.

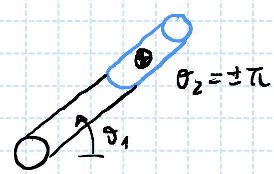
$$\begin{cases} \tau_1 = (4+\cos(\theta_2^*))\,\ddot{\theta}_1 \\ \tau_2^* = (1+\cos(\theta_2^*))\,\ddot{\theta}_1 + \dot{\theta}_1^2 \sin(\theta_2^*) \end{cases}$$, τ_2^* is the breaking torque.

(a) The perceived inertia is $B_{11} = 4+\cos(\theta_2^*)$

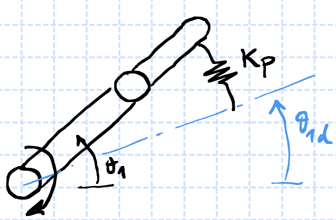
The maximum inertia is $B_{11,max} = 5$, when $\theta_2^* = 0$

The minimum inertia is $B_{11,min} = 3$, when $\theta_2^* = \pm\pi$



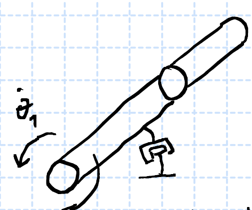


(b) PD controller, $\tau_1 = -K_d\,\dot{\theta}_1 - K_p(\theta_1 - \theta_{1d})$



$\tau_p = -K_p(\theta_1-\theta_{1d})$

↳ stiffness



$\tau_d = -K_d\dot{\theta}_1$

↳ damping

$\tau_1 = (4+\cos(\theta_2^*))\,\ddot{\theta}_1$, $(4+c_2^*)\,\ddot{\theta}_1 = -K_d\,\dot{\theta}_1 - K_p(\theta_1 - \theta_{1d})$, (closed-loop dynamics)

$$\Rightarrow \underbrace{(4+c_2^*)}_{B_{11}}\ddot{\theta}_1 + K_d\,\dot{\theta}_1 + K_p\,\theta_1 = K_p\,\theta_{1d}$$

Laplace Transform $\Rightarrow (B_{11}s^2 + K_d s + K_p)\,\Theta_1(s) = K_d\,\Theta_{1d}(s)$

$$\boxed{F_{II}(s) = \frac{\omega_n^2}{s^2+2\xi\omega_n s+\omega_n^2}}$$

$\omega_n \rightarrow$ frequency

$\xi \rightarrow$ damping ratio

$$\frac{\Theta_1(s)}{\Theta_{1d}(s)} = \frac{K_p}{B_{11}s^2 + K_d s + K_p} = \frac{K_p/B_{11}}{s^2 + \frac{K_d}{B_{11}}s + \frac{K_p}{B_{11}}},$$

$$\omega_n^2 = \frac{K_p}{B_{11}} \Rightarrow \omega_n = \sqrt{\frac{K_p}{B_{11}}},$$

$$2\zeta\omega_n = \frac{K_d}{B_{11}} \Rightarrow \zeta = \frac{1}{2}\sqrt{\frac{B_{11}}{K_p}}\frac{K_d}{B_{11}} = \frac{1}{2}\frac{K_d}{\sqrt{K_p B_{11}}}.$$

$\omega_{n,max}$, ζ_{max} whenever $B_{11,min}$, or $\theta_2 = \pm\pi$

$\omega_{n,min}$, ζ_{min} whenever $B_{11,max}$, or $\theta_2 = 0$.

(2) $B\ddot{q} + \phi + g = \tau$, $q = \begin{pmatrix} \theta_1 \\ d_2 \\ d_3 \end{pmatrix}$

$$B = \begin{pmatrix} \bar{B}_{11} + m_3 d_3^2 & 0 & 0 \\ 0 & m_2+m_3 & 0 \\ 0 & 0 & m_3 \end{pmatrix}, \quad \phi = \begin{pmatrix} 2m_3 d_3 \dot{\theta}_1 \dot{d}_3 \\ 0 \\ -m_3 d_3 \dot{\theta}_1^2 \end{pmatrix}, \quad g = \begin{pmatrix} 0 \\ (m_1+m_3)|g| \\ 0 \end{pmatrix}$$

$$\bar{B}_{11} = I_{z,1} + I_{y,2} + I_{y,3}$$

- Gravity compensation, The dynamics due to gravity are removed through

$$\tau = \tau_c + \hat{g}.$$

$\hat{g}$: estimate

$$B\ddot{q} + \phi + g = \tau_c + \hat{g} \underset{g=\hat{g}}{\Rightarrow} B\ddot{q} + \phi = \tau_c.$$

Assumption 1: The gravity vector is accurately known, $g = \hat{g}$.

- Linearization for the maximum inertia configuration,

$$B_{11,max} = \bar{B}_{11} + m_3 d_{3,max}^2 \Rightarrow B_{max} = \begin{pmatrix} B_{11,max} & 0 & 0 \\ 0 & B_{22} & 0 \\ 0 & 0 & B_{33} \end{pmatrix}$$

Assumption 2: $d_{3,max}$ is the maximum displacement for joint 3,

$$0 < d_{3,min} < d_3 < d_{3,max}$$

Assumption 3: The velocities are very low, so that $\phi \approx 0$.

$$B_{max}\ddot{q} = \tau_c \quad \text{(linearized system)}$$

- PD Controller,

$$\tau_{c_i} = -K_{d_i}\dot{q}_i - K_{p_i}(q_i - q_{id})$$

$$\omega_{n_i} = \sqrt{\frac{K_{p_i}}{B_{ii}}}, \quad \zeta_i = \frac{K_{d_i}}{2\sqrt{B_{ii}K_{p_i}}}$$

Assumption 4: Critical damping, $\zeta_i = 1$, and equal frequencies $\omega_{n_i} = \omega_n$ are good choices for the controller's design.

$$K_{p_i} = \omega_n^2 B_{ii} \; , \quad K_{d_i} \underset{\zeta=1}{=} 2\sqrt{B_{ii}\,\omega_n^2 B_{ii}} = 2\omega_n B_{ii} \; .$$

$$\tau_{c_i} = -K_{d_i}\dot{q}_i - K_{p_i}(q_i - q_{id}) \qquad B_{max} = \begin{pmatrix} B_{11,max} & 0 & 0 \\ 0 & B_{22} & 0 \\ 0 & 0 & B_{33} \end{pmatrix}$$

$$\tau_c = -2\omega_n B_{max}\dot{q} - \omega_n^2 B_{max}(q - q_d)$$

$$\tau = \tau_c + \hat{g} \; .$$

- Nonlinearities, Following assumptions 1 and 3, the nonlinear effects are due only to the change in the inertia at joint 1.

$$B(d_3)\ddot{q} + 2\omega_n B_{max}\dot{q} + \omega_n^2 B_{max} q = \omega_n^2 B_{max} q_d$$

$$B = \begin{pmatrix} \bar{B}_{11} + m_3 d_3^2 & 0 & 0 \\ 0 & m_2 + m_3 & 0 \\ 0 & 0 & m_3 \end{pmatrix}$$

$$\bar{B}_{11} = I_{z,1} + I_{y,2} + I_{y,3}$$

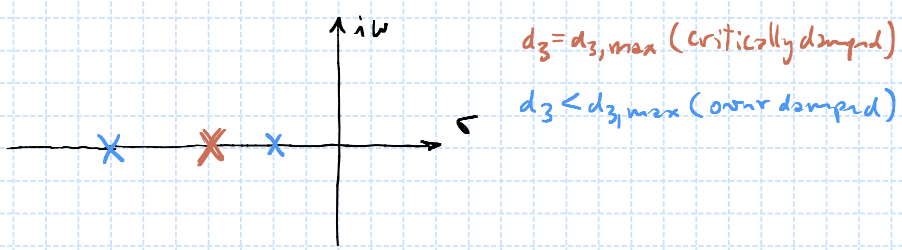
joint 1, $\dot{q} \approx 0$, $\hat{g} = g$

$$B_{11}\ddot{\theta}_1 + 2\omega_n B_{11,max}\dot{\theta}_1 + \omega_n^2 B_{11,max}\theta_1 = \omega_n^2 B_{11,max}\theta_{1d}$$

$$\Rightarrow \ddot{\theta}_1 + \underbrace{\frac{2\omega_n B_{11,max}}{B_{11}}}_{2\zeta_a \omega_a}\dot{\theta}_1 + \underbrace{\frac{\omega_n^2 B_{11,max}}{B_{11}}}_{\omega_a^2}\theta_1 = \underbrace{\frac{\omega_n^2 B_{11,max}}{B_{11}}}_{\omega_a^2}\theta_{1d}$$

$$F_{II}(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$\omega_a = \omega_n\sqrt{\frac{B_{11,max}}{\bar{B}_{11} + m_3 d_3^2}} \; , \quad 2\zeta_a\omega_a = \frac{2}{\omega_n}\omega_a^2 \Rightarrow \zeta_a = \frac{\omega_a}{\omega_n} = \sqrt{\frac{B_{11,max}}{\bar{B}_{11} + m_3 d_3^2}} \; .$$



$d_3 \downarrow \Rightarrow \omega_a \uparrow, \zeta_a \uparrow$

//

In the project, you can determine the maximum inertia moments (along the diagonal of B) through visual inspection. The result, to design the decentralized controller, will be a diagonal matrix, B_{max}, containing the previous values.